# Paramita Debbarma

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## EXPERIENCE

**1M1B Green Internship** **India**
*Climate Tech & AI Intern* May 2025 - June 2025

- Built **Greenie Chat**, an AI-powered sustainability assistant using Flask, Langchain, RAG, Llama-3 and Pinecone, trained on 100+ curated climate resources.
- Developed retrieval workflows, context handling and prompt engineering to improve response accuracy and relevance by 30%.
- Researched and organized 50+ climate and renewable energy concepts into a structured knowledge base.
- Applied Design Thinking, SWOT and SMART frameworks to plan and evaluate sustainability solutions.

**Flaunch Emerging Tech** **India**
*Generative AI Intern* October 2024

- Completed a 4-week intensive program focused on Generative AI and Large Language Models (LLMs), including LLaMA models.
- Gained hands-on experience in AI, machine learning and prompt engineering through 5+ guided assignments and model experiments.
- Built a **Personalized Learning Assistant**, leveraging Llama models for AI-driven learning support.
- Program conducted in partnership with AICTE and 1M1B Foundation, selected among national-level participants.

**Adverk Education** **India**
*Artificial Intelligence Intern* April 2024 - June 2024

- Gained hands-on experience in AI, machine learning, Data Science and Python-based analytics.
- Performed EDA on a 48k+ row dataset(Adult Income) using Python, NumPy and Pandas.
- Implemented ML models including Logistic Regression, Decision Tree, Random Forest and XGBoost.
- Evaluated model performance using accuracy scores, confusion matrices and R-squared metrics; created insights using Power BI.

## SKILLS

Programming & Frameworks: Python, JavaScript, Flask, Streamlit, TensorFlow, Keras, HTML/CSS, MySQL.
AI/ML & Analytics: Generative AI, RAG, LLMs, Groq API, NumPy, Pandas, Scikit-learn, Statistical Analysis.
Others: Git/github, Design Thinking, Problem Solving, Team Collaboration, SWOT & SMART Frameworks.

## PROJECT

**Synthetic Tabular Data Generation & Evaluation for Agri Crop Recomm** **|B.Tech, Final Year Project**

- Developed a unified evaluation framework to assess synthetic agricultural data across 4 dimensions: statistical fidelity, multivariate structure preservation, downstream ML utility and empirical privacy risk.
- Implemented 5 generative models - CGAN, CVAE, WGAN-GP, CGAN and TVAE and assessed them using 9 evaluation metrics on one soil and crop recommendation dataset.
- Built an interactive Streamlit application for evaluation, visualization and composite model ranking.

**Flood Detection System**

- A Streamlit-based application that identifies flood-affected areas using Sentinel-1 SAR GRD data and Google Earth Engine. The system compares pre and post-flood satellite imagery with threshold-based classification to generate flood maps. Visual results are displayed interactively, with options to export GeoTIFFs. Built with Python libraries including Rasterio, NumPy, Matplotlib, and OpenCV.

## EDUCATION

**National Institute of Technology, Delhi** **Narela, Delhi - 110040**
Master of Technology, Mathematics and Computing (2026- present)
**Tripura Institute of Technology** **Agartala, Tripura(West)**
Bachelor of Technology, Computer Science and Engineering(2022-26) Aggregate CGPA: 8.16

## ADDITIONAL EXPERIENCE AND ACHIEVEMENTS

- Completed the 1M1B 4-Day Workplace Experience Sprint, gaining hands-on exposure to teamwork, rapid prototyping and real workplace problem-solving while building **InSync** in <48 hours.
- Showcased my project - **Greenie Chat** as a Young Changemaker at the 1M1B Green Skills @ 2047 event hosted at UN India, New Delhi and delivered a talk on AI-driven climate solutions and responsible technology use to an audience of 50+ delegates and leaders.